```
use ass10

db.students.drop()

db.students.insertMany([
  { _id: 1, name: "Amit", age: 15, marks: 85, city: "Delhi" },
  { _id: 2, name: "Priya", age: 14, marks: 90, city: "Mumbai" },
  { _id: 3, name: "Ravi", age: 16, marks: 78, city: "Kolkata" },
  { _id: 4, name: "Simran", age: 17, marks: 88, city: "Bangalore" },
  { _id: 5, name: "Arjun", age: 15, marks: 92, city: "Delhi" },
  { _id: 6, name: "Meera", age: 16, marks: 81, city: "Mumbai" },
  { _id: 7, name: "Karan", age: 17, marks: 75, city: "Kolkata" },
  { _id: 8, name: "Sonia", age: 14, marks: 95, city: "Bangalore" },
  { _id: 9, name: "Vikram", age: 16, marks: 80, city: "Chennai" },
  { _id: 10, name: "Anjali", age: 15, marks: 89, city: "Chennai" }
])

db.students.createIndex({ city: 1 })

db.students.find().pretty()

db.students.aggregate([
  {
    $group: {
      _id: null,
      averageMarks: { $avg: "$marks" }
    }
  }
])

db.students.aggregate([
  {
    $group: {
      _id: "$city",
      totalStudents: { $sum: 1 }
    }
  }
])

db.students.aggregate([
  { $sort: { marks: -1 } }
])

db.students.aggregate([
  {
    $match: { city: { $in: ["Delhi", "Mumbai"] } }
  },
  {
    $group: {
      _id: "$city",
      averageMarks: { $avg: "$marks" }
    }
  }
])
```

```
// ------------------------------------------------------------
// 📘 COMMENT SECTION (FOR REPORT / VIVA)
// ------------------------------------------------------------
//
// 🔹 DATABASE USED: sagar_ass10
// 🔹 COLLECTION: students
// 🔹 NUMBER OF RECORDS: 4
//
// ------------------------------------------------------------
// 🔹 OBJECTIVE:
// To understand MongoDB's aggregation framework and how indexes
// can improve query performance.
// ------------------------------------------------------------
//
// 🔹 AGGREGATE OPERATORS USED:
// ▪ $group → Groups documents and performs aggregate functions (like $sum, $avg)
// ▪ $avg → Calculates average value
// ▪ $sum → Counts or totals documents
// ▪ $sort → Orders results
// ▪ $match → Filters documents before grouping
//
// ------------------------------------------------------------
// 🔹 INDEX USED:
// ▪ createIndex({ city: 1 }) → Creates ascending index on the 'city' field
//   Improves search efficiency for city-based queries.
//
// ------------------------------------------------------------
// 🔹 EXPECTED OUTPUT SUMMARY:
//
// ➤ Average marks of all students → 85.25
// ➤ Total students per city → 1 each
// ➤ Sorted students by marks → Priya (90), Simran (88), Amit (85), Ravi (78)
// ➤ Average marks per city (Delhi, Mumbai) → Delhi: 85, Mumbai: 90
//
// ------------------------------------------------------------
// 🔹 VIVA SHORT ANSWERS:
//
// Q1️⃣: What is aggregation in MongoDB?
// ➤ Aggregation is a process of transforming and combining data
//   using operations like $group, $sum, $avg, $sort, etc.
//
// Q2️⃣: What is the purpose of $group in aggregation?
// ➤ $group is used to group documents by a specific field and
```

```
//    perform calculations like sum, average, or count.
//
// Q 3 : What is the difference between $match and find()?
// ➤ $match filters data *inside an aggregation pipeline*, while
//    find() retrieves documents directly.
//
// Q 4 : What is an index in MongoDB?
// ➤ An index is a special data structure that improves the speed of
//    data retrieval operations on a collection.
//
// Q 5 : How do you check if an index is created?
// ➤ Run: db.students.getIndexes()
//
```